

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)**(TRIMESTER –IV) END TERM EXAMINATION, OCTOBER-2023****LOGISTICS MANAGEMENT & INFORMATION SYSTEM****[Time: 2 Hours 30 Minutes]****[Maximum Marks: 70]**

[**BL** – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)
CO – Course Outcomes]

Following are the course outcomes of the subject: - Logistics Management & Information System

CO	Course Outcome (CO)
CO1	Understand the nature of management information systems and their applications in business.
CO2	Use information management process for business value.
CO3	Critically understand the information management process & develop decision support skill.
CO4	Know the important business functions provided by typical business softwares.

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)**

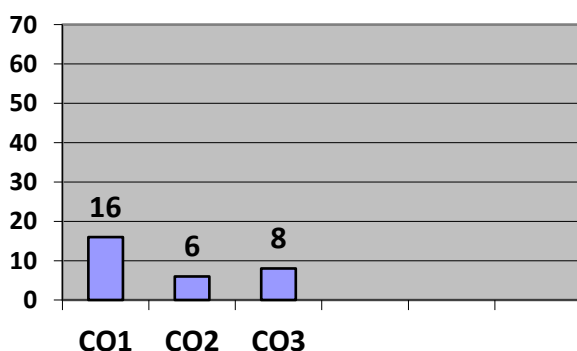
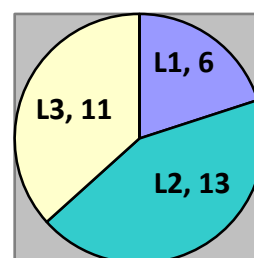
Q.1	Marks	CO	BL
A	4	CO - 1	L - 1
B	4	CO - 1	L - 1
C	4	CO - 2	L - 2
D	4	CO - 3	L - 2
E	4	CO - 4	L - 1

SECTION-B**Attempt any 3 questions out of the following: (Long Answer Type Questions)**

Q. No	Marks	CO	BL
2 (A)	8	CO - 1	L - 2
2 (A)	8	CO - 1	L - 2
2 (B)	8	CO - 2	L - 3
2 (B)	8	CO - 2	L - 3
3 (A)	8	CO - 3	L - 3
3 (A)	8	CO - 3	L - 3
3 (B)	8	CO - 4	L - 1
3 (B)	8	CO - 4	L - 1

SECTION-C**Attempt all parts of the following: (Caselet / Numerical)**

Question	Marks	CO	BL
4 (A)	9	CO - 3	L - 3
4 (B)	9	CO - 3	L - 2

Course Outcome Wise Marks Distribution**Bloom's Level wise Marks Distribution****Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2022-24)

(TRIMESTER – IV) END TERM EXAMINATION, October - 2023

LOGISTICS MANAGEMENT & INFORMATION SYSTEM

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 70]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) Discuss the fundamental role and importance of information systems in business.
- (b) How do you distinguish between process control and inventory control in MIS (management information system)?
- (c) Illustrate the concept of transaction system in MIS for a manufacturing organization.
- (d) Do accuracy, reliability, objectivity and conciseness of data and information management make an organization agile? Explain citing examples.
- (e) List down the challenges of collecting and analysing data.

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(4×8=32)

- (a) Classify the components of management information system (MIS) in an organization. Also discuss any five general characteristics of MIS.

OR

- (a) What are the major requirements for office automation in a logistics company?
- (b) Compare and contrast business intelligence vs. business analytics.

OR

- (b) Suggest what kind of integrated information system a global automaker will need to address supply chain issues. Explain.

Q.3. (a) How do technologies create business value for an organization? Discuss any five recent strategic technology trends.

OR

- (a) In what way the following add value to an organization?

- (i) Voice based order picking system
- (ii) Transaction processing systems
- (iii) Automatic identification technology
- (iv) Electronic point of sale (EPOS)

- (b) Examine a traditional warehouse and an automated warehouse performance on the overall business performance of FMCG organization.

OR

- (b) Write short notes on:

- (i) Electronics data interchange (EDI)
- (ii) Enterprise resource planning (ERP)

SECTION-C

Q.4. Attempt all parts of the following:

(2×9=18)

RFID IMPLEMENTATION AT ABHISHEK INDUSTRIES

RFID enables the company to gain better visibility and traceability of work in progress

The Ludhiana based Abhishek Industries is one of the leading textile exporters, exporting to over 40 countries and supplying to the biggest brands in the retail market. Strict control over quality, therefore, becomes a given. The use of RFID technology, in conjunction with ERP, has helped the company drive greater operational efficiencies, eventually aiding in quality adherence.

Problem Areas

Abhishek Industries faced the challenge of tracking semi-finished goods in greige folding at its Terry Towel Plant. It was critical for the company to track the Work in Progress to bring in the required operational efficiencies in the manufacturing process.

The greige stock area has running towels wound on very large pipes, called doffs. This is the stage between the weaving and dying stage. At any given point, there are around 1,000 doffs lying in the area. According to Ravi Tandon, CIO, Abhishek Industries, “Manual tracking of the doffs is a very cumbersome task and also throws open various challenges like locating the right doff. Though the doffs look similar, they may belong to different orders and have different characteristics and quality. A lot of time was lost in trying to locate the right doff as someone had to manually go through. There is also the probability of the doff coming in last and going out first, as that would be the easiest thing to do for the person handling it manually. So, ensuring the right sequence of first in, first out was becoming difficult.”

RFID to the rescue

The company outlined on the RFID solution from Avaana to automate the tracking of doffs. As part of the solution, active RFID tags were placed on each of the doffs, where RFID readers were placed in strategic locations in the greige stock area. The data with respect to a particular doff is fed into the tag, which is then read by RFID reader. As a result, the system can help the floor manager know the stock levels; and where are the physical stocks, whether there are any non-moving stocks. It also sends alarm signals when unauthorized doffs are moved.

Tags have an up-to-date and real time status of doffs at any given point of time. So, if material has been taken from doffs and used, the tag will provide the latest position at that point of time. The company has integrated its RFID system with its SAP R/3 ERP System and ensured that the technology works consistently. The company invested around Rs 25 lakh on the RFID implementation. The implementation, however, was not without its set of challenges as the implementation partners had to work around the interference with metals. There was specific casing done for each tag, depending on which material it was going to be placed on.

The RFID implementation has given the company clear visibility of the WIP, something it did not have before. Today, the company is able to determine the current position of the doff and is able to ensure its first in first out usage. This has led to saving of time earlier lost in trying to locate which doff is lying where. Further, there has been an improvement in the operational efficiencies as the loader is now able to pick up the correct doff for the specific order. All this eventually helps in maintaining the required

Future Plan

Abhishek Industries has outlined plans to extend the usage of the technology to other areas within the company as well. This includes areas like finished goods warehouse, cotton warehouse for raw material tracking, tracking of certain critical movable assets, and manpower access control and attendance. The pilot project for employee attendance recording and access control through RFID has already been done.

4(a) Implementing RFID was Abhishek industries’ own decision. Was it a part of global strategy? Explain your answer in detail.

4(b) What are the key challenges that the Abhishek Industries faced while implementing RFID technology? List down the key benefits that the company achieved by implementing RFID.